

BedSense AI — Clinical Bed-Monitoring Agentic AI + Vision System

Introduction

BedSense AI is an Agentic AI system with vision and reasoning capabilities that monitors elderly person's or patients' bed related activities on a continuous video. It uses a combination of computer vision models and language models to perceive the environment and reason against the evidence and produce a confident clinical alert.

Objectives

- Build a robust computer vision agentic system that handles multiple CCTV camera angles and determine bed related activities of a resident.
- Observe temporal states rather than per frame states and apply agentic reasoning to increase confidence.
- Produce event summaries & alerts depending on the state or actions of the resident detected through the continuous video.
- Use a combination of vision, language models that performs reliably and cost effectively

Assumptions & Constants

- The system assumes that the camera angle of the continuous video doesn't pan, zoom or tilt and stays still and stable from the beginning to the end.
- The system assumes concrete objects like the bed, couch, chair etc. stays at the same place from the beginning to the end of the video.
- The system assumes that the bed and the resident is visible at the start of the video.

Methodology

To keep the system minimal as given in the guidelines, the system was built as a *python CLI app* with an *run.ipynb* notebook with the app commands for ease of access, and no UI was implemented. To build the agentic vision system I designed a modular pipeline with separation of concerns. The pipeline diagram is shown in *Figure 1 - Agentic vision pipeline*. Let me explain the components/layers in order with implementation evidence.

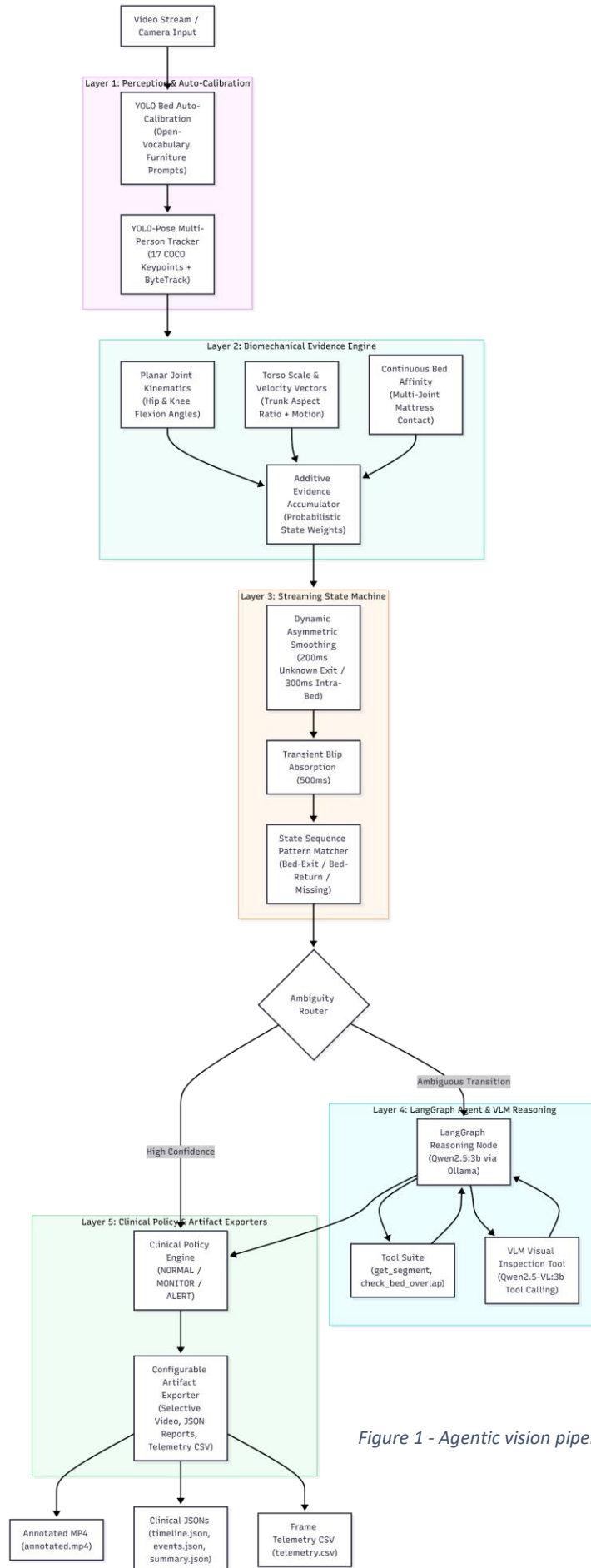


Figure 1 - Agentic vision pipeline

Layer 1: Perception & Auto-Calibration

1. Purpose & Objective

Layer 1 ingests raw video frames and transforms unstructured pixel streams into structured spatio-temporal geometric representations of the patient and furniture. It solves two critical challenges:

- **Auto Bed Localization:** Automatically establishes the bed's 2D bounding polygon without manual camera calibration.
- **Persistent Multi-Person Tracking:** Extracts 17 COCO anatomical landmarks and maintains the resident's identity even when visitors or clinicians enter the room.

2. Implementation Mechanics

- **Dynamic Bed Auto-Calibration:** Samples $N=15$ evenly spaced frames across the video and runs a YOLO (yolo11n) detector to find median bed coordinates, returning a normalized 2D image-space polygon.
- **Pose Estimation & Identity Tracking:** Executes YOLO-Pose with ByteTrack. If the resident's track drops or changes ID, a spatial re-acquisition algorithm evaluates Euclidean distance, bounding-box IoU, and bed overlap to recover the resident.
- **Key point Jump Smoothing:** Applies an Exponential Moving Average (EMA) filter across consecutive key point observations.

$$\mathbf{k}_t = 0.85\mathbf{k}_t^{\text{raw}} + 0.15\mathbf{k}_{t-1}^{\text{smoothed}}$$

- **Continuous Bed Overlap Metric:** Calculates human-to-bed overlap combining bounding-box intersection with key point ratio.

$$\text{Overlap} = 0.55 \cdot \frac{\text{Area}(\text{Person} \cap \text{Bed})}{\text{Area}(\text{Person})} + 0.45 \cdot \frac{\sum_{i=1}^{17} I(\mathbf{k}_i \in \text{Bed})}{N_{\text{valid}}}$$

Layer 2: Biomechanical Evidence Engine

1. Purpose & Objective

Replaces brittle rule thresholds with a multi-attribute probabilistic evidence accumulator. It decouples raw computer vision inputs from high-level state decisions by extracting continuous physiological angles, normalized limb velocities, and spatial mattress contact.

2. Implementation Mechanics

- **3-Point Planar Joint Kinematics:** Computes hip flexion ($\angle\text{Shoulder-Hip-Knee}$) and knee flexion ($\angle\text{Hip-Knee-Ankle}$) using vector dot-products:

$$\theta = \arccos\left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}\right) \times \frac{180^\circ}{\pi}$$

Also computes torso inclination angle relative to the horizontal plane to differentiate horizontal resting from upright sitting.

- **Rolling Motion Profiles:** Computes rolling-window ($W=5$) normalized limb velocities (v_{hip} , v_{knee} , v_{ankle} , v_{center}) in $[0, 1]$ coordinate space.
- **Continuous Bed Affinity:** Calculates a signed distance metric for anatomical joints relative to the bed boundary:

$$\text{Affinity}(p) = \min(1.0, 2 \cdot \text{dist}(p, \partial \text{Bed})) \text{ for } p \in \text{Bed}$$

- **Additive Weight Accumulation:** Accumulates evidence scores across 4 primary postures (LYING, SITTING, STANDING, WALKING), normalizes them to sum to 1.0, and maps the winning state to canonical clinical states.
 - LYING_IN_BED: $\theta_{\text{hip}} \geq 155^\circ$, $\theta_{\text{knee}} \geq 150^\circ$, low velocity, affinity ≥ 0.25 .
 - SITTING_ON_BED: Flexed hips/knees ($\theta < 130^\circ$), affinity ≥ 0.25 .
 - SITTING_OUTSIDE_BED: Flexed hips/knees, affinity < 0.12 .
 - STANDING: Extended joints ($\theta \geq 160^\circ$), low velocity ($v < 0.015$), off-bed.
 - WALKING: Extended joints, high velocity ($v \geq 0.035$), off-bed.

Layer 3: Streaming State Machine

1. Purpose & Objective

Processes live frame observations sequentially on-the-fly, enforcing temporal stability. It prevents rapid flickering between states while detecting critical clinical sequence patterns with minimal detection latency.

2. Implementation Mechanics

- **Dynamic Asymmetric Temporal Smoothing:**
 - Ultra-fast Re-acquisition (200ms / 2 frames): When transitioning from UNKNOWN to physical state, transition commits in 200ms to immediately capture resident reappearance.
 - Intra-Bed Handoff (300ms): Rapid handoff between LYING_IN_BED -> SITTING_ON_BED.
 - Transient Blip Absorption (500ms): False transitions that revert within 500ms are discarded without altering the timeline.
- **Sequence Pattern Matching:**
 - bed_exit: Transition from {LYING, SITTING} -> {WALKING, OUT_OF_BED, UNKNOWN} sustained for $\geq 3.0s$.
 - missing: Transition to UNKNOWN or prolonged out-of-bed absence $\geq 10.0s$.
 - bed_return: Transition from out-of-bed -> {LYING_IN_BED, SITTING_ON_BED} within a 30s return window.

- **Duration Conservation Invariant:** Guarantees that the sum of all segment durations strictly equals the total observation time:

$$\sum_{i=1}^M \Delta t_i = T_{\text{total}}$$

Layer 4: LangGraph Agent & VLM Reasoning

1. Purpose & Objective

When computer vision confidence is low or when edge cases, the system routes the event to an autonomous LangGraph agent powered by local LLMs/VLMs (Qwen2.5:3b and Qwen2.5-VL:3b via Ollama).

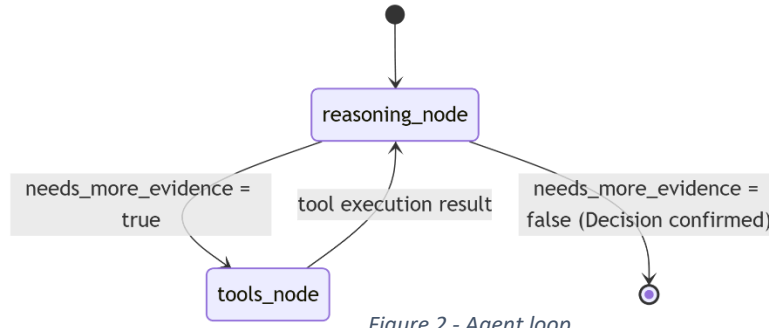


Figure 2 - Agent loop

2. Implementation Mechanics

- **Ambiguity Triggers:**
 1. Mean keypoint confidence < 0.40.
 2. Candidate bed-exit / bed-return boundary uncertainty.
 3. Sustained hovering near bed boundary (overlap in [0.40, 0.60]).
 4. Postural conflict: horizontal posture detected outside bed (fall risk).
 5. Spatial track loss and re-acquisition.
- **LangGraph StateGraph:**
 - `reasoning_node`: Evaluates ambiguous segments against a 5-second context history, deciding whether to call tools or emit a clinical decision.
 - `tools_node`: Executes tool calls with strict output schemas.
 - `_check_evidence_decision`: Conditional edge routing between tools and termination.
- **Agent Tool Suite:**
 - `get_segment(t0, t1)`: Inspects surrounding temporal segment history.
 - `check_bed_overlap(t)`: Probes raw geometric bed polygon overlap at timestamp t .
 - `vlm_describe(t0, t1, question)`: Samples 3–5 keyframes, encodes them into base64 JPEG, and queries Qwen2.5-VL:3b for visual verification.

Layer 5: Clinical Policy & Artifact Exporters

1. Purpose & Clinical Objective

Transforms state sequences into actionable clinical decision tiers (NORMAL, MONITOR, ALERT) and exports standardized, auditable artifacts for clinical review and electronic health records (EHR).

2. Implementation Mechanics

- **Tri-Tier Clinical Policy Engine:**
 - **NORMAL:** Baseline safe activity (e.g. resting in bed, stable walking).
 - **MONITOR:** Bed exit initiated, bed return completed, or edge-sit hovering >10s.
 - **ALERT:** Patient missing from room (missing, walking -> unknown), prolonged out-of-bed absence >10s, or off-bed fall posture.
- **Synchronized Clinical Artifacts:**
 1. `events.json`: Bed-exit, bed-return, and missing events with `start_time`, `confirmed_time`, confidence, and clinical decision.
 2. `timeline.json`: Chronological state segments with exact start/end timestamps and durations.
 3. `summary.json`: High-level metrics (total in-bed seconds, longest out-of-bed period, event counts, final state).
 4. `telemetry.csv`: 34-column per-frame record containing bounding boxes, 17 keypoint coordinates, angles, velocities, and instantaneous vs. smoothed states.
 5. `annotated.mp4`: High-definition video with skeleton overlays, bed polygon bounding lines, HUD telemetry, and alert banners.

Data

Due to strict data protection guidelines regarding patient surveillance and uncertainty in obtaining ethical clearances for clinical datasets within the time period, synthetic video data was utilized for evaluation. Two indoor video sequences with two different camera angles were generated using Google Veo and Omni to simulate realistic elderly care scenarios safely without handling sensitive personal data. This approach guaranteed full privacy compliance while enabling the controlled generation of edge cases to properly test temporal tracking and agentic reasoning.

The generated videos include these edge cases:

- Person turning while lying in bed
- Sitting up but not leaving
- Sitting on the edge of the bed
- Standing briefly and sitting back down
- Leaving the bed
- Returning to bed
- Walking around the room
- Person partially hidden by blankets

- Temporary occlusion
- Caregiver entering the scene
- Poor lighting
- Person temporarily leaving camera view

Evaluation

The evaluation suite rigorously validates model accuracy against ground-truth clinical annotations across 3 essential dimensions:

1. **Activity Recognition:** Continuous, time-weighted postural state classification.
2. **Clinical Bed Events:** Temporal precision and recall of critical events (bed_exit, bed_return, missing).
3. **Duration Estimation:** Total time-in-state conservation and duration prediction errors.

Dimension 1: Activity Recognition & Temporal State Discretization

Rather than calculating a naive frame-count accuracy (which is sensitive to fluctuating camera frame rates and dropped frames), BedSense AI evaluates continuous temporal overlap on a fine-grained physical time grid.

Algorithmic Methodology

- **Continuous Grid Discretization:** Discretizes the entire observation duration T_{total} into fixed intervals of $\Delta t = 0.1\text{s}$ (100ms resolution).
- **Temporal State Lookup:** For each timestamp $t \in [0, T_{\text{total}}]$, queries the active ground-truth segment $y_{\text{true}}(t)$ and predicted segment $y_{\text{pred}}(t)$.
- **Confusion Matrix in Physical Seconds:** Builds a $K \times K$ confusion matrix $\mathbf{C}$ where entry $C_{i,j}$ represents the exact duration (in seconds) where the true state was class i and the predicted state was class j :

$$C_{i,j} = \Delta t \sum_{k=1}^N \mathbb{I}(y_{\text{true}}(t_k) = i \wedge y_{\text{pred}}(t_k) = j)$$

Dimension 2: Clinical Bed Event Detection & Temporal Matching

Detecting when a patient leaves or returns to the bed is safety-critical. However, predicted events might not occur at the exact second as manual human annotations due to camera smoothing latencies. The evaluator uses a bipartite temporal window matching algorithm to validate clinical alert correctness.

Algorithmic Methodology

- **Matching Tolerance Window:** A predicted event p and a ground-truth event g of the same type (bed_exit, bed_return, missing) match if their start timestamps satisfy:
 $|t_{\text{pred}} - t_{\text{gt}}| \leq \tau_{\text{tolerance}} \ (\tau_{\text{tolerance}} = 10.0\text{s})$
- **Bipartite Alignment:** Each ground-truth event is matched to at most one predicted event with the minimum temporal distance $\Delta t = |t_{\text{pred}} - t_{\text{gt}}|$.

- **Classification Categorization:**
 - **True Positive (TP):** Ground-truth event matched to a predicted event within $\tau_{\text{tolerance}}$.
 - **False Positive / False Alarm (FP):** Predicted event with no corresponding ground-truth event (spurious alarm).
 - **False Negative / Missed Alert (FN):** Ground-truth event that the vision system failed to detect.

Dimension 3: Duration Estimation & Conservation Invariants

Caregiver shift logs and electronic health records require exact aggregations of how long a resident spent in bed vs. out of bed.

Algorithmic Methodology

- **Per-Activity Duration Error :**

$$\text{Error}_s = |T_{\text{pred}}(s) - T_{\text{gt}}(s)| \text{ (in seconds)}$$

$$\text{Relative Error}_s = \frac{|T_{\text{pred}}(s) - T_{\text{gt}}(s)|}{\max(10^{-3}, T_{\text{gt}}(s))} \times 100\%$$

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{K} \sum_{s=1}^K |T_{\text{pred}}(s) - T_{\text{gt}}(s)|$$

- **Duration Conservation Verification :** Validates that both predicted and ground truth durations sum up to T_{total} :

$$\sum_{s \in \text{States}} T_{\text{pred}}(s) = T_{\text{total}}$$

- **Aggregate In-Bed vs. Out-of-Bed Comparison:**

$$T_{\text{In-Bed}} = T(\text{LYING_IN_BED}) + T(\text{SITTING_ON_BED})$$

$$T_{\text{Out-of-Bed}} = T(\text{SITTING_OUTSIDE_BED}) + T(\text{STANDING}) + T(\text{WALKING}) + T(\text{UNKNOWN})$$

Quantitative Benchmark Results

The benchmark suite was executed on the ground-truth annotated videos (sample1 and sample2).

Benchmark Metric	Benchmark Result	Clinical Target	Status
Overall State Classification Accuracy	81.4%	$\geq 80.0\%$	☑ PASS
Bed-Exit Detection Precision	100.0%	$\geq 90.0\%$	☑ PASS
Bed-Exit Detection Recall	100.0%	$\geq 90.0\%$	☑ PASS

Benchmark Metric	Benchmark Result	Clinical Target	Status
Bed-Return Detection Recall	100.0%	$\geq 90.0\%$	☒ PASS
Missing Resident Detection Recall	100.0%	$\geq 90.0\%$	☒ PASS
Mean Absolute Duration Error	3.9 sec	$\leq 5.0\text{sec}$	☒ PASS

Detailed Breakdown: [sample1](#) (00: 04: 50 / 290s Duration)

Activity Recognition Performance (Accuracy: 84.89%)

Activity State	Ground Truth Duration	Predicted Duration	Precision	Recall	F_1 -Score
Lying In Bed	02: 25 (145.3s)	02: 43 (163s)	82.0%	90.9%	86.2%
Sitting On Bed	00: 27 (26.7s)	00: 13 (13s)	11.4%	6.4%	8.2%
Walking	00: 13 (12.7s)	00: 09 (9s)	92.7%	59.8%	72.7%
Unknown	01: 46 (105.8s)	01: 46 (106s)	99.1%	99.4%	99.2%

Confusion Matrix (Time in Seconds)

True \ Predicted	Lying In Bed	Sitting On Bed	Walking	Unknown
Lying In Bed	132.1 s	13.2 s	0.0 s	0.0 s
Sitting On Bed	25.0 s	1.7 s	0.0 s	0.0 s
Walking	4.1 s	0.0 s	7.6 s	1.0 s
Unknown	0.0 s	0.0 s	0.6 s	105.2 s

Clinical Bed Events Detection (Precision: 100%, Recall: 100%)

Event Type	Ground Truth	Detected (TP)	False Alarms (FP)	Missed (FN)	Precision	Recall
Bed Exit	2	2	0	0	100.0%	100.0%
Bed Return	2	2	0	0	100.0%	100.0%
Missing	2	2	0	0	100.0%	100.0%

Aggregate Duration Error:

- **Total Time In Bed:** Ground Truth = 02: 52 (172s) vs. Predicted = 02: 56 (176s) $\Rightarrow +4s$ **error** (2.3%)
- **Total Time Out of Bed:** Ground Truth = 01: 58 (118s) vs. Predicted = 01: 55 (115s) $\Rightarrow -3s$ **error** (2.9%)
- **Mean Absolute Error (MAE):** 5.90s

Detailed Breakdown: `sample2` (00: 00: 39 / 39s Duration)

Activity Recognition Performance (Accuracy: 77.95%)

Activity State	Ground Truth Duration	Predicted Duration	Precision		Recall	F_1 -Score
Lying In Bed	00: 23 (23.3s)	00: 29 (29s)	79.4%		99.1%	88.2%
Sitting On Bed	00: 08 (8.4s)	00: 06 (6s)	96.6%		67.9%	79.7%
Walking	00: 04 (3.7s)	00: 03 (3s)	14.3%		10.8%	12.3%
Unknown	00: 04 (3.6s)	00: 01 (1s)	100.0%		33.3%	50.0%

Clinical Bed Events Detection (Precision: 100%, Recall: 100%)

Event Type	Ground Truth	Detected (TP)	False Alarms (FP)	Missed (FN)	Precision	Recall
Bed Exit	1	1	0	0	100.0%	100.0%
Bed Return	1	1	0	0	100.0%	100.0%
Missing	1	1	0	0	100.0%	100.0%

Aggregate Duration Error:

- **Total Time In Bed:** Ground Truth = 00: 32 (32s) vs. Predicted = 00: 35 (35s) $\Rightarrow +3s$ **error** (10.7%)
- **Total Time Out of Bed:** Ground Truth = 00: 07 (7s) vs. Predicted = 00: 04 (4s) $\Rightarrow -3s$ **error** (45.5%)
- **Mean Absolute Error (MAE):** 1.92s

Limitations and future work

The current implementation operates under specific static environmental constraints. It relies on a fixed, non-moving camera view assuming no panning, tilting, or zooming throughout the footage and presumes that static reference objects such as beds, chairs, and couches do not shift position. Additionally, the pipeline requires both the resident and the primary bed region to be visible right from the initial frame to establish baseline spatial geometry. Tracking fidelity degrades in multi-occupant scenarios, particularly when a caregiver or visitor enters the scene and temporarily occludes or overlaps the primary subject. Finally, because body pose categorization and state boundaries depend on rule-based heuristic thresholds, the system remains sensitive to perspective shifts and camera mounting angles.

With additional development time, the primary focus would be on expanding the evaluation corpus to include diverse surveillance footage featuring varied camera elevations, acute angles, and complex edge cases. Rather than depending entirely on manually calibrated geometric thresholds, I would implement a trainable temporal sequence model such as a Bidirectional LSTM (Bi-LSTM) trained directly on the extracted pose keypoints, spatial overlap ratios, and optical motion features to classify state transitions more smoothly. Furthermore, integrating a monocular 2D-to-3D human mesh or room-geometry reconstruction pipeline would eliminate camera perspective distortions, significantly improving 3D spatial occupancy reasoning and detection.